

Rupjyoti Barman

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Summary

.NET Backend Developer with 3+ years of experience building scalable APIs and system-level diagnostic platforms for **HP client systems**. Strong expertise in **C#, ASP.NET Core**, and managed-unmanaged interoperability, with experience in designing **decoupled architectures** and performance-critical backend systems.

Technical Skills

- **Languages:** C#, C++, SQL
- **Frameworks:** ASP.NET Core (**Web API, MVC**), Entity Framework Core
- **Technologies:** REST APIs, WMI, WUA API, HPCMSL, Hardware Diagnostics
- **Architecture:** Clean Architecture, Dependency Injection, Factory Pattern, RPC
- **Tools:** Git, SQL Server, PostgreSQL

Experience

Software Developer, BoldTek India Pvt. Ltd. – Guwahati
Client: Hewlett-Packard (HP)

June 2023 – Present

- Designed and developed centralized hardware diagnostic APIs using **C# (.NET)**, serving as a shared backend consumed by multiple HP client applications; implemented **layered architecture** with native **C++ DLL** integration to perform diagnostics across disk (**SMART, Read/Verify**) and network (**NIC IRQ, ROM validation**), ensuring reliable system validation.
- Led architectural redesign by migrating from a shared **RPC-based service** model to a decoupled **NuGet-based solution**; consolidated business logic, API layer, and diagnostic modules into a unified package enabling local execution and reducing failures caused by multi-team service instability.
- Built a real-time hardware monitoring module using **WMI (ManagementEventWatcher)** with **WQL event subscriptions** on **Win32_PnPEntity** to detect device insertion and removal; implemented an **event-driven mechanism** to notify UI components.
- Developed diagnostic modules for detecting driver updates via **Windows Update Agent (WUA) API** and BIOS update checks using **HPCMSL**, enabling user-driven update actions.
- Developed and enhanced an **ASP.NET Core MVC** application for managing INI-based configurations of HP NPI systems; implemented **CRUD operations, role-based access control**, and validation using **Entity Framework Core** and **SQL Server**.
- Collaborated with cross-functional global teams (India, Taiwan, US) to deliver features and resolve production issues.

Projects

Ticket Management System (ASP.NET Core Web API)

- Developed a production-ready backend using **ASP.NET Core Web API** following **SOLID principles** and **clean architecture**; designed data models using **Entity Framework Core** with **PostgreSQL** and implemented **JWT-based authentication**.
- Built scalable **REST APIs** for ticket lifecycle management using **DTO-based design**, validation, and **dependency injection**; implemented logging using **Serilog**.

Education

B.Tech in Computer Science & Engineering - CGPA: 8.01/10
North Eastern Regional Institute of Science and Technology (NERIST), Arunachal Pradesh

May 2023